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标题	Diastereoselective Formal 1,3-Dipolar Cycloaddition of Trifluoroethyl Amine-Derived Ketimines Enables the Desymmetrization of Cyclopentenediones
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摘要（原文节选）

In this research, a metal-free diastereoselective formal 1,3-dipolar cycloaddition of N-2,2,2-trifluoroethylisatin ketimines and cyclopentene-1,3-diones which can efficiently lead to the desymmetrization of cyclopentene-1,3-diones is developed. With the developed protocol, a series of tetracyclic spirooxindoles containing pyrrolidine and cyclopentane subunits can be smoothly obtained with good results (up to 99% yield and 91:9 dr). Furthermore, the methodology can be extended to trifluoromethyl-substituted iminomal ⋯

数据表 · Table 1（13 行，含数值行 13 行）

Optimization of reaction conditions [a].

Entry	Base	Solvent	Time (h)	dr (%) [b]	Yield [c]
1	TMG	DCE	12	81:19	73
2	DBU	DCE	12	77:23	87
3	DBN	DCE	12	69:31	59
4	TBD	DCE	12	77:23	66
5	CS2CO3	DCE	96	71:29	30
6	DBU	toluene	12	73:27	58
9	DBU	THF	12	74:26	42
10	DBU	MeCN	12	70:30	79
11	DBU	MeOH	12	64:36	73
12	DABCO	DCE	48	84:16	90
13 [d]	DABCO	DCE	24	84:16	90
14 [d,e]	DABCO	DCE	24	84:16	92
15 [d,e,f]	DABCO	DCE	24	85:15	96

脚注（逐字）：[a] Unless otherwise noted, the reaction was carried out with 1a (0.12 mmol), 2a (0.1 mmol), and 1.0 equivalent of base in 1.0 mL of solvent at 30 ° C. [b] Determined by 1H NMR analysis of the crude product. [c] Yield of isolated product as a mixture of diastereoisomers. [d] Run at 50 ° C. [e] The rat

正文对应句（逐字）：Initially, we selected the N-2,2,2-trifluoroethylisatin ketimine 1a and cyclopentene-1,3-dione 2a as model substrates for the optimization of reaction conditions. As shown in Table 1, to our delight, when using 1,1,3,3-tetramethylguanidine (TMG) as the base the reaction could proceed well in 1,2-dic

数据表 · Table 2（13 行，含数值行 13 行）

Substrate scope investigation of metal-free diastereoselective 1,3-dipolar cycloaddition of N-2,2,2-trifluoroethylisatin ketimines 1a and diverse cyclopentene-1,3-diones 2 [a].

Entry	2 (R2/R3)	3	dr [b]	Yield (%) [c]
1	2b (Me/3-MeC6H4)	3ab	84:16	96
2	2c (Me/4-MeC6H4)	3ac	82:18	94
3	2d (Me/3-MeOC6H4)	3ad	85:15	94
4	2e (Me/4-FC6H4)	3ae	88:12	96
5	2f (Me/2-ClC6H4)	3af	71:29	76
6	2g (Me/4-ClC6H4)	3ag	86:14	94
7	2h (Me/3-BrC6H4)	3ah	84:16	84
8	2i (Me/4-BrC6H4)	3ai	85:15	94
9	2j (Me/1-naphthyl)	3aj	89:11	98
10	2k (Me/2-naphthyl)	3ak	86:14	99
11	2l (Et/C6H5)	3al	88:12	69
12	2m (Et/3-BrC6H4)	3am	72:28	84
13	2n (Et/4-BrC6H4)	3an	84:16	88

脚注（逐字）：[a] The reactions were carried out with 1a (0.15 mmol), 2 (0.10 mmol), and 1.5 equivalents of DABCO in 1.0 mL of DCE at 50 ° C for 24 h. [b] Determined by 1H NMR analysis of the crude product. [c] Yield of isolated product as a mixture of diastereoisomers.

正文对应句（逐字）：Initially, we selected the N-2,2,2-trifluoroethylisatin ketimine 1a and cyclopentene-1,3-dione 2a as model substrates for the optimization of reaction conditions. As shown in Table 1, to our delight, when using 1,1,3,3-tetramethylguanidine (TMG) as the base the reaction could proceed well in 1,2-dic

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